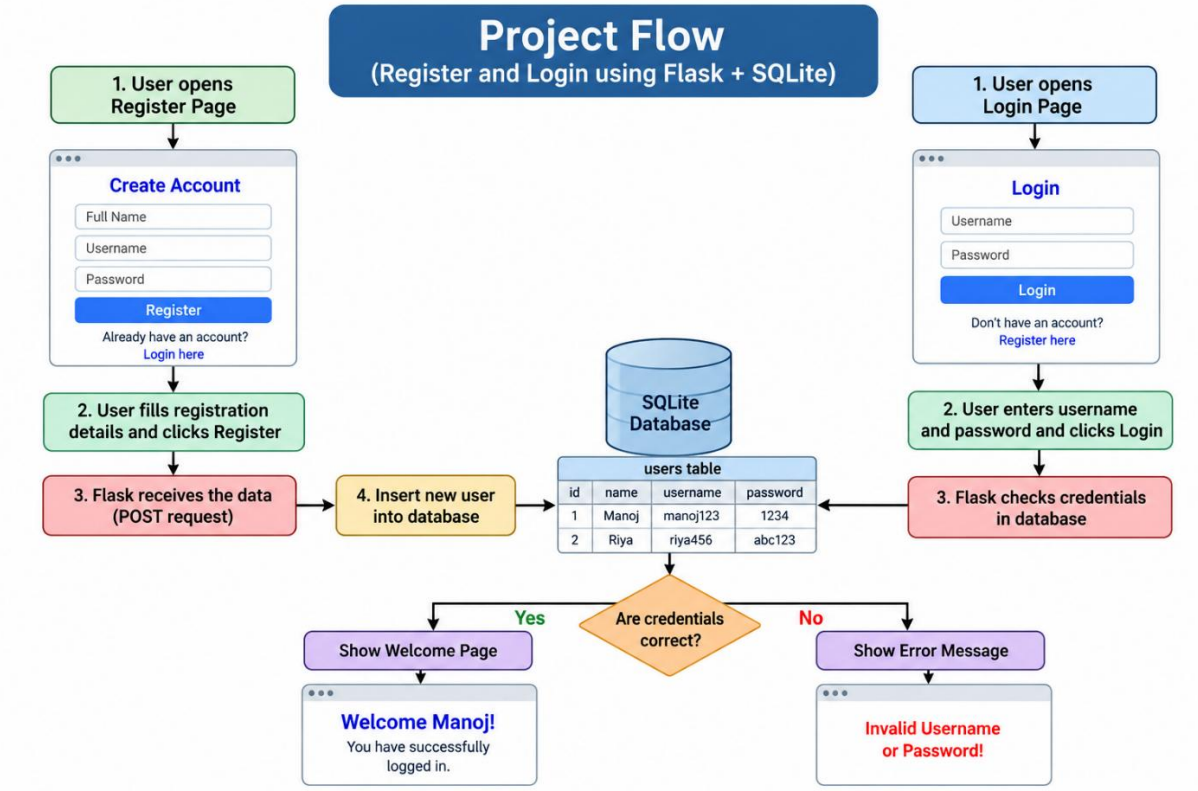
# LOGIN FORM USING FLASK




## 1. Project Structure

login_app/

```
│
├── app.py
│
└── templates/
    ├── login.html
    ├── register.html
    └── welcome.html
```

## 2. Flask Backend with Database

**app.py**

```
from flask import Flask, render_template, request
import sqlite3
app = Flask(__name__)
```

```
def create_database():
    conn = sqlite3.connect("database.db")
    cursor = conn.cursor()
    cursor.execute("""
        CREATE TABLE IF NOT EXISTS users (
            id INTEGER PRIMARY KEY AUTOINCREMENT,
            name TEXT NOT NULL,
            username TEXT NOT NULL UNIQUE,
            password TEXT NOT NULL
        )
    """)
    conn.commit()
    conn.close()

@app.route("/", methods=["GET", "POST"])
def login():
    if request.method == "POST":
        username = request.form["username"]
        password = request.form["password"]
        conn = sqlite3.connect("database.db")
        cursor = conn.cursor()
        cursor.execute(
            "SELECT * FROM users WHERE username = ? AND password = ?",
            (username, password)
        )
        user = cursor.fetchone()
        conn.close()
        if user:
            return render_template(
                "welcome.html",
                name=user[1]
            )
        else:
            return "Invalid Username or Password!"
```

```
    return render_template("login.html")

@app.route("/register", methods=["GET", "POST"])
def register():
    if request.method == "POST":
        name = request.form["name"]
        username = request.form["username"]
        password = request.form["password"]
        conn = sqlite3.connect("database.db")
        cursor = conn.cursor()
        try:
            cursor.execute(
                """
                INSERT INTO users (name, username, password)
                VALUES (?, ?, ?)
                """,
                (name, username, password)
            )
            conn.commit()
            conn.close()
            return """
                <h2>Registration Successful!</h2>
                <a href="/">Login here</a>
            """
        except sqlite3.IntegrityError:
            conn.close()
            return """
                <h2>Username already exists!</h2>
                <a href="/register">Try another username</a>
            """
    return render_template("register.html")
if __name__ == "__main__":
    create_database()
    app.run(debug=True)
```

## 3. Login HTML

**templates/login.html**

```
<!DOCTYPE html>
<html>
<head>
  <title>Login</title>
<style>
  body {
    background-color: #f2f6ff;
    font-family: Arial, sans-serif;
    text-align: center;
    padding-top: 100px;
  }

  h1 {
    color: #2563eb;
    font-size: 35px;
  }

  input {
    padding: 10px;
    margin: 8px;
    width: 250px;
    font-size: 16px;
  }

  button {
    padding: 10px 25px;
    background-color: #2563eb;
    color: white;
    border: none;
    font-size: 16px;
    cursor: pointer;
```

```
    }

    p {
        color: #555;
    }

    a {
        color: #2563eb;
        text-decoration: none;
    }
</style>
</head>
<body>
<h1>Login</h1>
<form method="POST">
    <input type="text"
           name="username"
           placeholder="Username"
           required>
    <br>
    <input type="password"
           name="password"
           placeholder="Password"
           required>
    <br><br>
    <button type="submit">Login</button>
</form>
<p>
    Don't have an account?
    <a href="/register">Register here</a>
</p>
</body>
</html>
```

## 4. Register Page

**templates/register.html**

```
<!DOCTYPE html>
<html>
<head>
  <title>Register</title>
<style>
  body {
    background-color: #f2f6ff;
    font-family: Arial, sans-serif;
    text-align: center;
    padding-top: 80px;
  }

  h1 {
    color: #2563eb;
    font-size: 35px;
  }

  input {
    padding: 10px;
    margin: 8px;
    width: 250px;
    font-size: 16px;
  }

  button {
    padding: 10px 25px;
    background-color: #2563eb;
    color: white;
    border: none;
    font-size: 16px;
    cursor: pointer;
```

```
    }

    p {
        color: #555;
    }

    a {
        color: #2563eb;
        text-decoration: none;
    }
</style>
</head>
<body>
<h1>Create Account</h1>
<form method="POST">
    <input type="text"
           name="name"
           placeholder="Full Name"
           required>
    <br>
    <input type="text"
           name="username"
           placeholder="Username"
           required>
    <br>
    <input type="password"
           name="password"
           placeholder="Password"
           required>
    <br><br>
    <button type="submit">Register</button>
</form>
<p>
    Already have an account?
```

```
    <a href="/">Login here</a>
</p>
</body>
</html>
```

## 5. Welcome Page

**templates/welcome.html**

```
<!DOCTYPE html>
<html>
<head>
    <title>Welcome</title>
<style>
    body {
        background-color: #f2f6ff;
        font-family: Arial, sans-serif;
        text-align: center;
        padding-top: 150px;
    }

    h1 {
        color: #2563eb;
        font-size: 40px;
        margin-bottom: 15px;
    }

    p {
        color: #555;
        font-size: 20px;
    }
</style>
</head>
<body>
<h1>Welcome {{ name }}! </h1>
```

You have successfully logged in.

</body>

</html>

## 6. Run the Python application

**Terminal → New Terminal**

i. PS C:\Users\YourName\Desktop\login_app>

ii. pip install flask

iii. python app.py

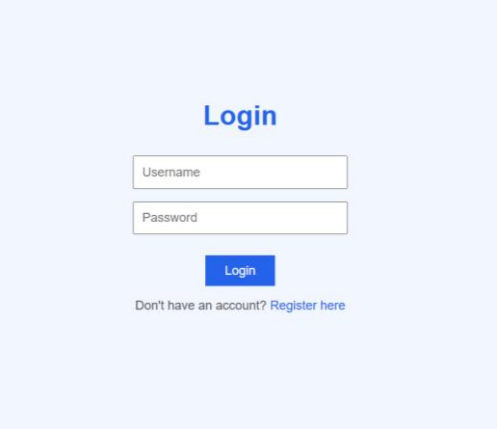
**You should get something similar to:**

```
* Serving Flask app 'app'
* Debug mode: on
* Running on http://127.0.0.1:5000
```

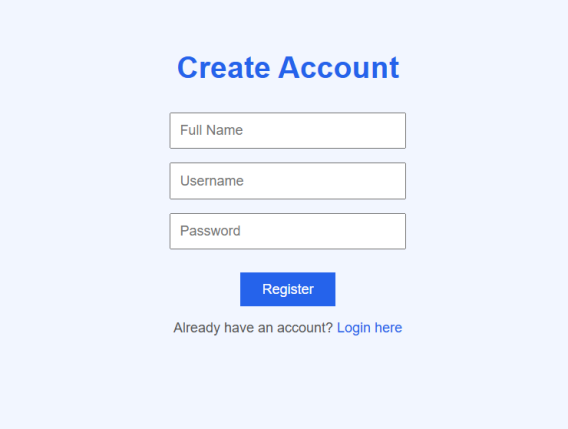
iv. Open the URL http://127.0.0.1:5000 that appears in the terminal.

## 7. Snapshot

**Login Page**




**Register Page**




**Welcome Page**

